Guardian: Technical Documentation

2026 Shenzhen FinTechathon · International Track · Topic A: Personal Finance Assistant

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1. Summary

Guardian is an AI assistant inside a bank's mobile app. It finishes money tasks for the client: it pays, cancels subscriptions, disputes charges, pays bills, builds and runs the budget, and saves toward goals. Next to it sits a bank kernel that lets the assistant do exactly what the client signed and nothing more.

The design rests on one rule: **models propose, code decides.** Agents (dispatcher, planner, readers) are untrusted. They send *intents* with references to where every value came from. A deterministic kernel then checks each intent:

- the agent's passport;
- the mandate the client signed;
- the provenance of every field;
- the bank's records;
- risk.

The kernel then either asks the client with a card it renders itself, or executes through a single-use grant. It checks the result with the bank or the merchant and writes everything to a signed, hash-chained journal.

PROMISE	MECHANISM
The agent finishes the job	Ten scenarios end to end on a sandbox bank; every result is verified from independent sources
The agent does exactly what you confirmed	Mandates in Mastercard Verifiable Intent format, signed by the client's device key; single-use grants; provenance labels
You never give the agent passwords or card numbers	Secret vault bound to the merchant domain; protected sign-in form; the LLM gateway refuses prompts with secrets
Repeat tasks are almost free	Model ladder; templates replay solved tasks with no model; the next month costs 63 % less than the first

Topic A asks for smart budgeting, bill analysis and spending analytics:

- **Budgeting** is scenario C8: auto-budget from history, envelopes, safe-to-spend, cash-gap forecast, salary split, goals.
- **Bill analysis** is C5 and C6: verification of every bill and detection of fake bills.
- **Spending analytics** is C3 and C4 (leaks, double charges) plus the month review.

The theme, *AI as a Financial Participant*, is the core of the design: the assistant acts, and the kernel makes acting safe.

The system is about 11 000 lines of Python, JavaScript and Solidity, with 3 350 lines of tests (346 tests, all green in CI with PostgreSQL and a local chain, plus 5 contract tests). It runs with `docker compose up` and needs no API keys in the default replay mode.

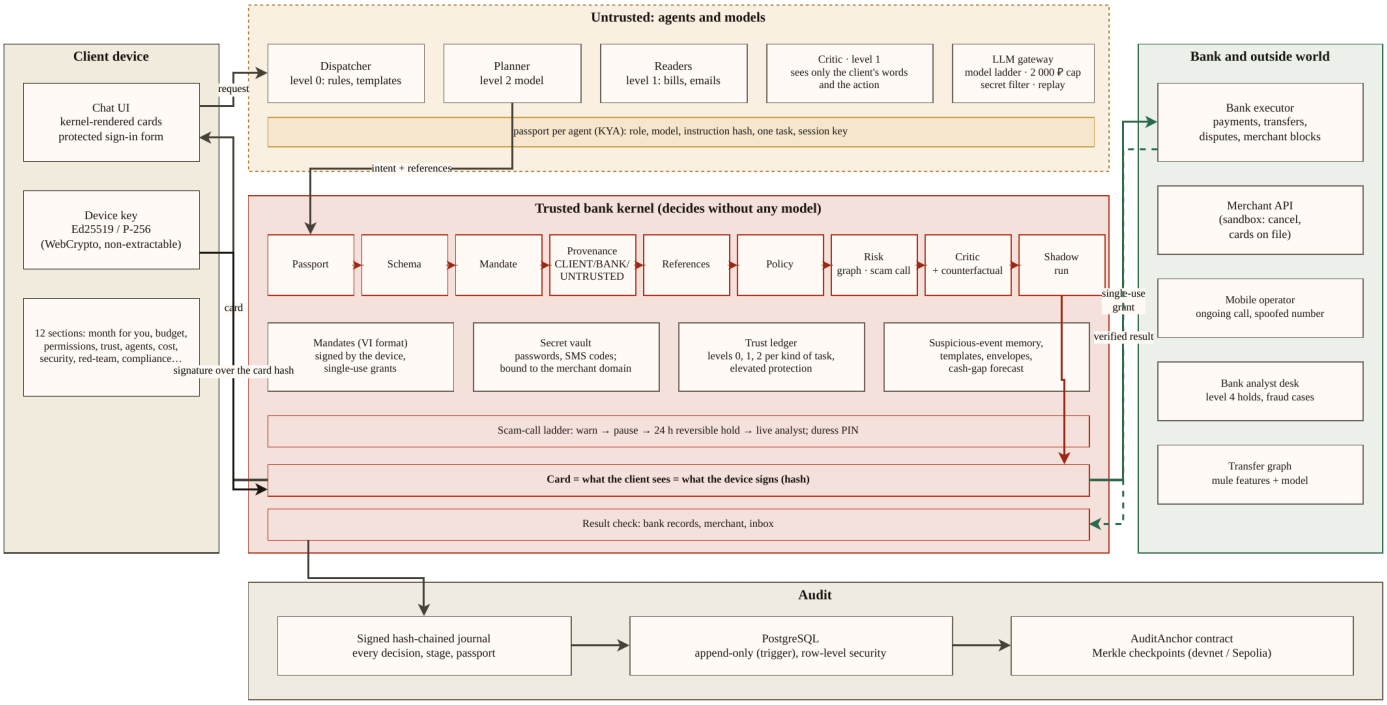
2. Scenarios

All data is synthetic. A generator builds one client with twelve months of history (1 156 transactions; seed 22; "now" is 17 Oct 2026 10:00 MSK). Brands, people and domains (`.example`) are fictional.

#	CLIENT SAYS	WHAT HAPPENS	VERIFIED BY
C1	"Pay for Melodify"	The planner builds an intent from kernel facts. The card shows payee, amount and source, all *verified by the bank*. The client signs once and can allow it monthly. Next month the payment runs from a template with no model and only a notification	Bank statement
C2	"Cancel the 399 ₽ subscription"	Two subscriptions match, so the kernel asks which. After the signature the bank blocks future charges from that merchant on that card (insurance). The cancellation desk then signs in on the merchant's official domain: credentials come from the bank vault, the SMS code is typed by the client. It cancels through the merchant API	Three sources: merchant removed the card on file; confirmation e-mail from the official domain; no charge when the bank ran time forward
C3	"Find leaks in my subscriptions"	Unused, duplicate and price-raised subscriptions, each with the transactions it is based on: 847 ₽ a month, 10 164 ₽ a year	Kernel analytics, 5/5 planted leaks found
C4	"Was I charged twice?"	The double charge is found with its two transaction ids; a dispute card follows	Dispute status in the bank, refund 3 days later
C5	"Pay my bills"	Readers extract fields from each bill. The kernel checks every field: the value occurs in the document, the account is in the payee directory, the sender domain is official, the amount fits history. Payment comes from the "Bills" envelope	Bank statement; bill marked paid
C6	(a fake bill in the inbox)	Look-alike sender, an account not in the directory, pressure words, hidden text for AI agents, and a link to an earlier suspicious e-mail: a case file for fraud analysts instead of a payment	Case opened, nothing paid
C7	(trial ends tomorrow)	Guardian asks before the first charge; cancelling runs C2 under a mandate that expires at the first charge	No charge after the trial
C8	"Make me a budget", "Save 60 000 by July, don't touch the gym", "Can I afford…"	Budget from 12 months, envelopes, safe-to-spend today, cash-gap warning with fixes from the client's own money (never credit), salary split on payday, goal plan, money freed by cancellations sent to the goal, month review, forecast accuracy	Back-test on held-out months
C9	"Transfer 350 000 ₽ to a safe account…" during a call	The telecom reports an ongoing call from a spoofed number. The kernel applies the scam-call ladder: explain → pause with three questions and PIN → 24-hour reversible hold with the trusted person told → live analyst. A duress PIN shows an ordinary receipt while nothing leaves	Hold record; the analyst's decision in the journal
C10	"Transfer 20 000 to account …"	The payee account is scored on the bank's own transfer graph. A mule-like collector raises the risk and the ladder level; an ordinary account is not flagged	Model evaluated on held-out graphs

`scripts/demo_run.py` runs the whole month (all ten scenarios, then one month later). It writes the readable log `evidence/demo_run.log` and the signed journal `evidence/demo_journal.jsonl`.

3. Architecture



Architecture: untrusted agents propose, the kernel decides, the bank executes, the journal records

There are four trust zones:

1. **Client device.**
   - Chat UI with cards rendered by the kernel.
   - A device key in WebCrypto: Ed25519, or ECDSA P-256 where Ed25519 is unavailable. The key is non-extractable.
   - The protected sign-in form for merchant credentials.
2. **Untrusted agents and models.**
   - Agents:
     - dispatcher: level 0 rules and templates, a level 1 model only if no rule fits;
     - planner: level 2;
     - readers: level 1, one document in, one structure out, no tools.
   - The critic (level 1) runs inside the kernel's pipeline but is a model, so it can only say "no" or "ask".
   - Every model call goes through the LLM gateway.
   - Every agent session holds a kernel-signed **passport**.
3. **Trusted kernel.** The only component that can move money. Its "allow" is always deterministic code.
4. **Bank and outside world.**
   - The bank executor (payments, transfers, disputes, merchant blocks).
   - Merchant APIs (sandbox).
   - The mobile operator's call signal.
   - The analyst desk.
   - The transfer graph.

Requests flow as follows:

1. The client's request reaches the dispatcher.
2. An agent proposes an intent with references.
3. The kernel resolves each reference itself and runs the pipeline.
4. The kernel renders a card.
5. The client's device signs the card's hash.
6. The kernel turns the signature into a mandate and issues a single-use grant.
7. The executor acts.
8. The kernel reads the bank back to confirm the result.

Every step is journaled with the autonomy stage and the passport of the agent involved.

4. The kernel

4.1 Decision pipeline

Every step can only tighten the verdict (allow → ask → deny). LLM-based steps plug in like the others but can only say "no" or "ask the client".

STEP	CHECKS	TYPICAL REASONS
Passport	Session signature, fresh nonce, role may call this endpoint, passport not expired, task matches	forged passport, wrong session key, replay, reader calling `propose_intent`
Schema	Action and fields are well formed; amounts are integers in kopecks	amount must be a positive integer in kopecks
Mandate	A signed mandate covers the action: payee in `allowed_payees`, amount in `amount_range`, budget and recurrence left, funding source, origin, not expired, not revoked	no mandate covers this action; already used this period
Provenance	Each field's label (CLIENT, BANK, UNTRUSTED) is assigned by the kernel from the reference, never by the model. Payee, amount and source must come from the client or the bank	payee comes from an unverified source
References	Every cited reference exists and resolves to the same value	reference to data that does not exist
Policy	Envelope balance, trust level, elevated protection, domain sets, no credit products, kill switch	envelope overspend; trust level 1 allows 1 000 ₽ alone
Risk	Payee directory, look-alike names, suspicious-event memory, usual amount, transfer-graph score, scam-call ladder	payee not in the directory; money mule (graph score 1.00)
Critic	A small model that sees only the client's words, the mandate and the action as structured fields with labels; cached; three refusals in a row return control to the client	2 990 ₽ is far above the usual 299 ₽
Counterfactual	On high risk only: the planner step is replayed with the client's request masked. If the same action reappears, it is driven by data rather than the client (after MELON, ICML 2025)	the agent proposes the same payment even without your request
Shadow	The payment is rehearsed on a copy of the bank: exactly one debit to the payee for the amount (or a move between the client's own accounts) and nothing else. A movement the bank would refuse goes back to the client	a dry run in a copy of the bank failed: insufficient funds; the dry run moved money differently from what was asked

After execution, `result.verified` reads the bank statement (or merchant account, or inbox) to confirm the effect. Only a verified result raises trust.

4.2 Mandates, cards and grants

A mandate is the exact right an agent gets after the client says “yes”. Field names follow Mastercard Verifiable Intent constraints. Our extensions are `allowed_origins`, `funding_source` and `secret_refs`.


```
{"id": "mdt_6f0263675996", "version": "vi-0.1", "client_id": "cl_1", "actions": ["pay"],
 "payment": {"allowed_payees": ["tok_melodify"],
             "amount_range": {"min": 0, "max": 34400, "currency": "RUB"},
             "budget": {"total": 103200, "currency": "RUB"},
             "agent_recurrence": {"frequency": "monthly", "max_occurrences": null},
             "reference": "7e47bb9b6f6d78c6…"},
 "allowed_origins": [], "funding_source": "card_1234", "secret_refs": [],
 "nbf": "2026-10-17T10:00:00+03:00", "exp": "2026-12-31T23:59:00+03:00",
 "standing": true, "single_use": false, "stage": 2, "parent_id": null, "holder": "client"}
```


The card for this mandate reads: *Pay Melodify up to 344 ₽, once a month, from card *1234, 1 032 ₽ in total, until 31 Dec 2026, 23:59* (amounts are kopecks in the mandate; `reference` is the hash of the card the client saw).

- **Card.** Rendered by the kernel from the mandate, never by a model. It lists what the agent will be able to do. It shows the provenance badge of every field and highlights what changed since last time. It carries the mandate hash.
- **Signature.** The device signs the hash. The kernel verifies it against the registered device key and countersigns. The card may require typing the amount (scam-call level 1+), answering questions and the PIN (level 2+).
- **Grant.** A single-use right for exactly one action, derived from the mandate. Rights only narrow: a child must fit inside its parent. A spent grant cannot be reused; a revoked mandate revokes its children.
- **Stages.** Operations are stage 1 (assisted: every payment through a card) or stage 2 (preset conditions: standing mandates with limits). Stage 3 (broad autonomy) is deliberately not supported, as the 2026 agent payment code allows.

4.3 Passports (Know Your Agent)

Each agent session gets a passport signed by the kernel. It contains:

- id, role, model and model level;
- the SHA-256 of the system instructions and the instruction-set version;
- the task id;
- issue and expiry times (one hour);
- an Ed25519 session key.

Requests are signed with the session key and carry a fresh nonce. Endpoints are allowed by role:

ROLE	ENDPOINTS
dispatcher	`route`, `lookup_templates`, `propose_intent`
planner	`propose_intent`, `query_data`, `request_grant`

ROLE	ENDPOINTS
reader	`submit_extraction`
critic	none (called by the kernel)
executor	`execute`

4.4 Earned trust

Trust is kept per kind of task (subscriptions, bills, own transfers, external payments, websites) and granted only from verified results:

- **Level 0:** a card for everything.
- **Level 1:** within a standing mandate the agent may act alone up to 1 000 ₽.
- **Level 2:** alone up to the mandate's limit.
- **Promotion** after 5 verified results in a row.
- **Demotion** one level on any kernel denial; down to 0 on a scope violation.
- **Elevated protection** (elderly, minors) pins external payments to level 0 and starts the scam-call ladder at the pause.

4.5 Secrets and cancellations (C2)

The vault stores merchant credentials encrypted with AES-GCM; the merchant's domain is part of the associated data. Agents see only references such as `{{secret:streambox.password}}`. Every plaintext is registered with the gateway's guard, so a leak into any prompt fails the call. SMS codes are one-time and destroyed after use.

The cancellation desk runs in five steps:

1. Block future charges from the merchant on the card (insurance).
2. Check the merchant's domain against the directory and look-alike detection.
3. Sign in through the protected form (credentials typed by the client go straight to the vault), or with saved credentials.
4. Cancel through the merchant API.
5. Verify from three sources.

The merchant's full card number is reduced to the last four digits at once.

4.6 Documents, suspicious memory and reference checks

- **Documents.** Readers return fields by schema. The kernel accepts a field only if:
  - it literally occurs in the document;
  - the account matches the directory;
  - the sender domain is official;
  - the amount fits the client's history.

  Accepted fields get the `document_verified` label and a reference agents may cite.
- **Injection screening.** Instructions aimed at AI agents and pressure tactics are detected in untrusted text. They never unlock anything; they only add reasons to stop.
- **Suspicious-event memory.** Look-alike senders, changed bank details and the like are remembered per client by the kernel only; agents cannot read or change this memory. This catches two-step attacks ("we changed our details" this week, a bill to the new details next week).

- **Reference check.** Models only phrase replies from kernel facts. Every number in a reply must be one the kernel supplied; otherwise the kernel's template is shown. Replies that offer credit, instalments or "pay later" are replaced unless the facts mention it.

5. Budget engine (C8)

All amounts are computed by deterministic code in `kernel/budget.py`; a model may only phrase the explanation.

- **Auto-budget.** Twelve months of history give income, mandatory payments, flexible categories and a savings target. Envelopes (Bills, Subscriptions, Food, Transport, Fun, Shopping, Health) become sub-balances of the current account.
- **Envelopes as a boundary.** An envelope is the `funding_source` of a mandate. The kernel never lets an agent pay beyond what the envelope holds, and an agent cannot move money between envelopes. Overspending gives the client a card with options.
- **Safe to spend today:**

$$\text{safe per day} = \frac{\text{balance} - \text{payments due before payday} - \text{goal transfers} - \text{cushion}}{\text{days to payday}}$$

- **Cash gap.** Salaries, recurring charges and bills are replayed day by day until the next income. A gap is shown with fixes from the client's own money only (savings, moving a payment, cutting an envelope), never a loan. Back-test: 1 of 1 gap predicted at least 3 days ahead, 0 false alarms in 18 probes.
- **Salary split** on payday: bills first, then subscriptions, goal and envelopes, with a note when an upcoming payment falls before the next salary.
- **"Can I afford it?"** The purchase is replayed against the forecast; the answer names the payment it would endanger.
- **Goals.** Example: 60 000 ₽ by 1 July 2027 means 6 700 ₽ from each of 9 salaries, with every month checked to keep bills covered and the untouchable category ("don't touch the gym") excluded from cuts. Money freed by cancellations can be added to the goal.
- **Month review and forecast quality.** Plan versus actual, spending spikes, late-night spending, price changes. Envelope forecasts for July to September 2026 were planned from the nine months before: WAPE is 27.3 % overall (bills 4.3 %, subscriptions 1.3 %, discretionary categories higher).

6. Payee risk on the transfer graph (C10)

- **Generator.** `risk_graph/generator.py` builds a synthetic intra-bank transfer graph (about 600 accounts per graph).
  - Normal life: salaries (legitimate fan-out), rent and shops (legitimate fan-in), friends and family, loans paid back.
  - Four planted mule schemes: ring, structuring below 15 000 ₽, fan-in/fan-out collector, chain through fresh accounts.
- **Features.** Fourteen per-account features computed only from transfers known at the scoring time:
  - account age;

    - in and out counts, unique counterparties, log volumes;
    - pass-through share, holding time;
    - transfers just below the threshold;
    - cycle membership, 48-hour bursts, regularity of outgoing amounts;
    - share of young peers.
- **Model.** Gradient boosting (scikit-learn), trained on graphs with seeds 101 to 103 and tested on graphs with seeds 201 to 205. Test accounts, victims and schemes were never seen in training.
- **Results on held-out graphs.** ROC-AUC 1.0, precision 0.993, recall 0.944, false-positive rate 0.0004, zero false positives on employers, merchants and landlords.
- **Recall by scheme.** Ring 0.972, structuring 0.907, fan-in/out 1.0, chain 0.893.
- **Caveat.** These are synthetic graphs from our own generator: the numbers show the pipeline works, not field accuracy.
- **Use.** A score of 0.7 or more adds a risk reason, asks the client and raises the scam-call ladder level. An unknown account gets no opinion.

7. Model ladder and cost

LEVEL	WHAT	USED BY
0	Kernel code, rules, templates	Dispatcher rules, templates, all decisions
1	Small model (Claude Haiku 4.5)	Readers, critic, counterfactual, reply wording
2	Medium model (Claude Sonnet 5.5)	Planner for new tasks
3	Large model, escalation only	Failed schema or low confidence at level 2

The LLM gateway is the only path to a model. It:

- holds the keys;
- starts at the cheapest level and escalates on facts (schema failure, low confidence, refusal);
- enforces a hard cap of 2 000 ₽ for the whole project plus a per-task budget;
- refuses prompts with card numbers or vault secrets;
- records responses for replay.

Modes: `replay` (default; falls back to deterministic stubs when no recording exists), `stub`, `record`, `live`.

Cost of one client's month, from `evidence/costs.json`, comes from token counts on the real prompts and stub answers priced at list prices. Live runs replace the estimate.

	MODEL CALLS	BIG-MODEL CALLS	TOKENS	COST
First month (seven tasks)	13	1	4 379	0.73 ₽
Next month (same tasks)	7	0	1 934	0.27 ₽ (63 % less)

The repeated Melodify payment and the repeated double-charge check use no model at all. At the next-month rate a million clients cost about 270 000 ₽ a month in model tokens. In the full demo run 96.7 % of the work happens at levels 0 and 1.

8. Data, persistence and anchoring

- **PostgreSQL** (enabled by `DATABASE_URL`; `docker compose` sets it).
  - Tables: `journal`, `mandates`, `vault_entries` (ciphertext only), `holds`, `cases`.
  - The application writes as the `guardian_app` role with the client set for the transaction, so row-level security applies to the application itself.
  - The journal is append-only: a trigger accepts only the next link of the hash chain, and every kernel run writes its own chain.
  - Migrations are in `db/migrations`; the tests run with `make test-pg`.
- **Journal.** Each entry has a sequence number, time, actor, event, data, autonomy stage, passport id, the previous hash, its own hash and an Ed25519 signature. `verify()` checks the chain and every signature.
- **Anchoring.** Merkle roots over entry hashes are written to the `AuditAnchor` contract (Solidity 0.8.30, Foundry, 5 contract tests): owner-only `anchor(chain, seq, root, head)`, with sequence numbers that only grow per chain.
  - The demo anchors four checkpoints on a local anvil devnet and reads them back.
  - The original journal matches. A rewritten-and-rehashed entry fails (Merkle root differs). A cut tail fails (the anchor covers more entries).
  - Only hashes, a sequence number and a run identifier go on chain.
  - Sepolia is supported by `make anchor` with an RPC and a funded test key.

9. Interfaces

Web app (`localhost:8000`):

- the chat with kernel cards, device signing, the protected sign-in form and receipts;
- twelve side sections: Month for you, Payments, Budget, Permissions (with one-tap revoke), Trust (with elevated protection), Agents at work, Cost, Security, Red-team, Compliance, Inbox, Journal;
- demo buttons that move the simulated clock and simulate a scam call.

HTTP APIs:

- **App** (`/api/…`): state, sections, chat, device registration, card confirm and decline, options, budget, mandates, cancellations, vault, time, demo call, protection, holds, journal, cases, server-sent events.
- **Kernel** (`/agent/…` with signed requests, client card endpoints, `/journal`, `/journal/verify`, `/journal/checkpoint`, `/operator/halt|resume` behind an operator token).
- **Sandbox bank** (HTTP rails with an executor token).

10. Testing and evaluation

SUITE	CONTENT	RESULT
Unit	Kernel modules, budget, analytics, crypto, journal, trust, graph features	pass
End-to-end	All scenarios through the service; the UI in Chromium (Playwright)	pass
Evaluation set	48 functional tasks (C1 to C10, cross-cutting; English, Russian, Chinese)	48 / 48

SUITE	CONTENT	RESULT
Attacks in the evaluation set	44 static and retention attacks + 6 adaptive goals	50 / 50 stopped
Red-team	39 static, 5 retention, adaptive search (360 attempts)	see the security report
Integration	PostgreSQL (RLS, append-only chain), anchoring on anvil	pass with `make test-pg`, Foundry
Contract	`forge test`	5 / 5

Reproduce:

- `make test` runs the test suites;
- `make evidence` produces all reports;
- `make anchor` runs the chain part.

The Docker image runs the whole suite at build time. A clean clone with the pinned requirements passes the suite and serves the app in replay mode without keys.

11. Deployment

```
docker compose up --build        # app + PostgreSQL 18, http://localhost:8000
```

- Python 3.14.
- Dependencies are pinned (`requirements.txt`).
- Configuration lives in `.env` (see `.env.example`): `GUARDIAN_LLM_MODE`, `ANTHROPIC_API_KEY` and `ANTHROPIC_BASE_URL` (live only), `GUARDIAN_TOKEN_BUDGET_RUB`, `DATABASE_URL`, `POSTGRES_PASSWORD`, `GUARDIAN_OPERATOR_TOKEN`.
- The model ladder is a values file, `config/models.yaml`: provider, model, prices and the effort flag for each level, plus the names of the variables that hold each provider's key and endpoint. Environment variables override it per level (`GUARDIAN_MODEL_L1`, `GUARDIAN_PROVIDER_L1`, `GUARDIAN_PRICE_L1_IN`, `GUARDIAN_PRICE_L1_OUT`, `GUARDIAN_EFFORT_L1`) or replace the whole file (`GUARDIAN_MODELS_FILE`). No model name and no key is written in the code.
- In a bank, the kernel, vault and executors run inside the bank's perimeter. Models are reached through the gateway only; a self-hosted model can replace a level without changing the kernel.

12. Ethics and responsible design

- **The assistant sells nothing.** It offers no credit, instalments or upgrades. Cash gaps are closed only with the client's own money. Credit offers in model text are replaced.
- **No pressure.** Cards are calm and factual. The assistant never uses urgency, and retention tricks from merchants (discounts, "pause instead", annual upsell) are refused by the kernel.
- **Fewer, better questions.** The client is asked once per new kind of action. Trust is earned from verified results, and standing permissions stay revocable with one tap.
- **Vulnerable clients.** Elevated protection, a trusted person, a duress PIN and a live analyst. The kernel never blocks accounts or calls the police on its own.
- **Privacy.** Models see structured facts, not raw statements, account numbers or secrets. Only hashes go on chain.

- **Explainability.** Every amount carries a calculation and the transactions behind it, and every decision carries its reasons in the journal.

13. Limitations

- The bank, merchants, telecom and transfer graph are sandboxes with synthetic data. Real data will be noisier, especially for the graph model.
- Cancellation uses merchant APIs. Merchants without an API would need a separate, strictly sandboxed integration, which we did not build.
- Costs are estimated on stub answers until live runs are recorded. The multi-model red-team (DeepSeek, Qwen) was not run without keys.
- Mandates are signed JSON. SD-JWT selective disclosure from Verifiable Intent is not implemented.
- Anchoring was demonstrated on a local devnet. Sepolia needs a funded test key.
- One client per demo world; scaling numbers are projections.

14. What’s next

- Pilot with a bank’s real payee directory and fraud labels.
- SD-JWT mandates.
- Merchant-side mandate verification (AP2 cart and payment mandates).
- A federated transfer-graph model across banks.
- Self-hosted small models for levels 0 and 1.
- Anchoring to a consortium chain.